

Cryptography and Steganography with Python



Steganography and cryptography have been widely used to hide sensitive information. This article offers a brief introduction to how both these methods can be used together. The implementation uses Python with the Stepic and ezPyCrypto libraries.

Cryptography is ‘the practice and study of hiding information.’ The word is a combination of the Greek *kryptos* (‘hidden’) and *gráphō* (‘I write’). Early cryptography was solely concerned with message confidentiality—making the message unreadable by interceptors, by encoding it with a *key*. Modern cryptography includes techniques for message integrity checking, sender/receiver identity authentication, digital signatures and more.

Algorithms that use the same key for both encryption and decryption are known as *symmetric key algorithms*. This practice, invented in the 1970s, uses a pair of keys—one to encrypt, the other to decrypt. Users of this technology publish their public key, while keeping their private key secret. This enables anyone to send them a message encrypted with the public key, which only the holder of the private key can decrypt. These algorithms are designed so that finding out the private